

ECE 2040: Homework 4

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Problem 4.5-2

The values of the mesh currents in the circuit shown in Figure P 4.5-2 are

$$i_1 = 2 \text{ A}, \quad i_2 = 3 \text{ A}, \quad i_3 = 4 \text{ A}$$

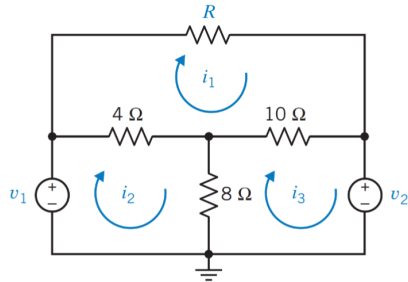
Determine the values of the resistance R and of the voltages v_1 and v_2 of the voltage sources.

Figure P 4.5-2

$$-v_1 + 4(i_2 - i_1) + 8(i_2 - i_3) = 0$$

$$8(i_3 - i_2) + 10(i_3 - i_1) + v_2 = 0$$

$$10(i_1 - i_3) + 4(i_1 - i_2) + i_1 R = 0$$

$$-v_1 + 4 - 8 = 0 \quad 8(1) + 10(2) + v_2 = 0$$

$$-v_1 - 4 = 0 \quad v_2 = -28 \text{ V}$$

$$v_1 = -4 \text{ V}$$

$$10(-2) + 4(-1) + 2R = 0$$

$$2R = 24$$

$$R = 12 \Omega$$

Problem 4.5-4

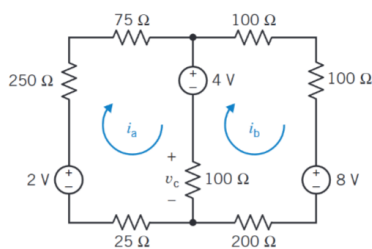
Determine the mesh currents, i_a and i_b , in the circuit shown in Figure P 4.5-4.

Figure P 4.5-4

$$-2 + 250i_a + 75i_a + 4 +$$

$$100(i_a - i_b) + 25i_a = 0$$

$$-4 + 100i_b + 100i_b + 8 + 200i_b$$

$$+ 100(i_b - i_a) = 0$$

$$2 + 450i_a - 100i_b = 0$$

$$450i_a - 100i_b = -2$$

$$500i_b - 100i_a = -4$$

$$4.5i_a - i_b = 20 \text{ mA}$$

$$-i_a + 5i_b = -40 \text{ mA}$$

Problem 4.5-5

Find the current i for the circuit of Figure P 4.5-5.

Hint: A short circuit can be treated as a 0-V voltage source.

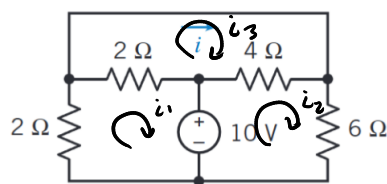


Figure P 4.5-5

$$2i_1 + 2(i_1 - i_3) + 10 = 0$$

$$-10 + 4(i_2 - i_3) + 6i_2 = 0$$

$$2(i_3 - i_1) + 4(i_3 - i_2) = 0$$

$$4i_1 - 2i_3 = -10 \quad i = i_3$$

$$10i_2 - 4i_3 = 10$$

$$-2i_1 - 4i_2 + 6i_3 = 0$$

$$\begin{bmatrix} 4 & 0 & -2 \\ 0 & 10 & -4 \\ -2 & -4 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \\ 0 \end{bmatrix}$$

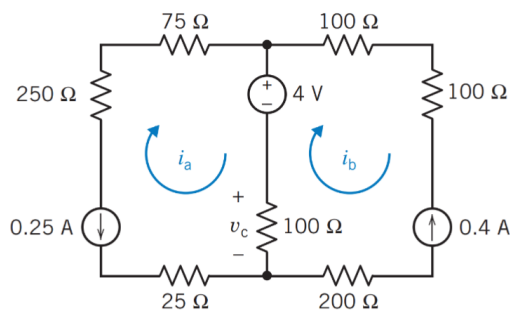


Figure P 4.6-2

$$i_a = -0.25 \text{ A}$$

$$i_b = -0.4 \text{ A}$$

$$V_c = (i_a - i_b) \cdot 100 \Omega$$

$$V_c = 15 \text{ V}$$

Problem 4.6-4

Find v_c for the circuit shown in Figure P 4.6-4.

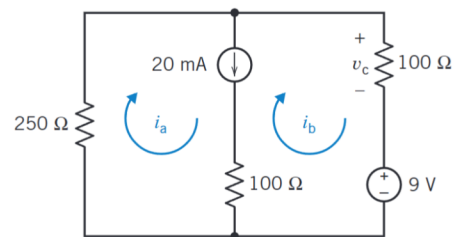


Figure P 4.6-4

$$i_a - i_b = 20 \text{ mA}$$

$$250i_a + 100i_b + 9 = 0$$

$$250(20 \text{ mA} + i_b) + 100i_b = -9$$

$$5 \text{ A} + 350i_b = -9 \text{ A}$$

$$350i_b = -14 \text{ A}$$

$$i_b = -40 \text{ mA}$$

$$V_c = 100 \Omega \cdot -40 \text{ mA}$$

$$i_a = -20 \text{ mA}$$

$$V_c = -4 \text{ V}$$

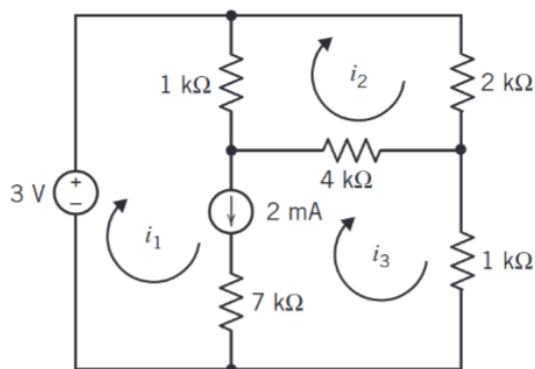


Figure P 4.6-8

$$4 \text{ k}\Omega (i_2 - i_3) + 1 \text{ k}\Omega (i_2 - i_1) + 2 \text{ k}\Omega (i_2) = 0$$

$$7 \text{ k}\Omega i_2 - 1 \text{ k}\Omega i_1 - 4 \text{ k}\Omega i_3 = 0$$

$$i_1 - i_3 = 2 \text{ mA}$$

$$-3 \text{ V} + 1 \text{ k}\Omega (i_1 - i_2) + 4 \text{ k}\Omega (i_3 - i_2) + 1 \text{ k}\Omega (i_3) = 0$$

$$1 \text{ k}\Omega i_1 - 5 \text{ k}\Omega i_2 + 5 \text{ k}\Omega i_3 = 3 \text{ V}$$

$$1 \text{ k}\Omega i_1 - 5 \text{ k}\Omega i_2 + 5 \text{ k}\Omega i_3 = 3 \text{ V}$$

Values of i_1, i_2, i_3
will be in mA.

$$\begin{pmatrix} 1 & 0 & -1 \\ 1 & -5 & 5 \\ -1 & 7 & -4 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$$

$$\frac{2-V_2}{4k\Omega} = i_a \quad 2-V_2 = 0.125mA \rightarrow 2-V_2 = 0.5V$$

$$V_2 = 1.5V$$

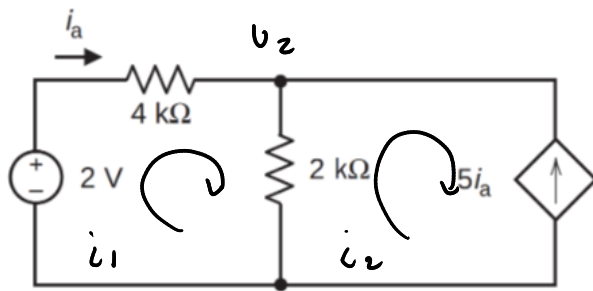


Figure P 4.7-2

$$-2V + 4k\Omega i_a + 2k\Omega (i_a + 5i_a) = 0$$

$$16k\Omega i_a = 2V$$

$$i_a = 0.125mA$$

$$CCCS = 5i_a = 0.625mA$$

$$p = vi = 2V \cdot 0.125mA = 0.25mW$$

$$p = vi = 1.5V \cdot 0.625mA = 0.9375mW$$

$$i_1 = i_a$$

$$i_2 = -5i_a$$

Problem 4.7-4

Determine the mesh current i_a for the circuit shown in Figure P 4.7-4.

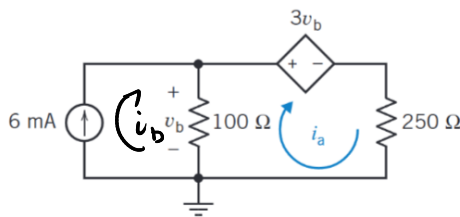


Figure P 4.7-4

$$3v_b + 250i_a + 100(i_a - i_b) = 0$$

$$350i_a - 100i_b = -3v_b$$

$$100(i_b - i_a) = v_b$$

$$v_b = 100i_b - 100i_a$$

$$-3v_b = 350i_a - 100i_b$$

$$v_b = .6 - 100i_a$$

$$-3v_b = 350i_a - .6$$

$$-1.8 + 300i_a = 350i_a - .6$$

$$-1.2 = 50i_a$$

$$i_a = -24mA$$

$$v_b = .6 + 2.4 = 3V$$

Problem 4.7-9

Determine the value of the resistance R in the circuit shown in Figure P 4.7-9.

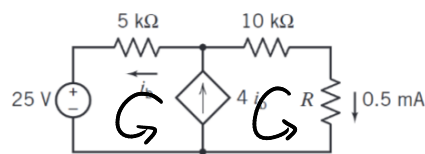


Figure P 4.7-9

$$i_b - i_a = 4i_b$$

$$i_a = -3i_b$$

$$i_b = 0.16mA$$

$$-0.5mA \cdot R + 10k\Omega \cdot i_a + 5k\Omega \cdot i_b + 25V = 0$$

$$-0.5mA \cdot R + 10k\Omega (-0.5mA) + 5k\Omega \cdot 0.16mA = -25V$$

$$-0.5mA \cdot R - 5V + 0.83V = -25V$$

$$\rightarrow R = 41.67k\Omega$$

Problem 4.7-15

Determine the values of the mesh currents i_1 and i_2 for the circuit shown in Figure P 4.7-15.

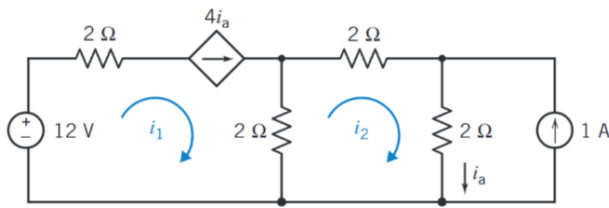


Figure P 4.7-15

$$-12V + 2i_1 + 2(i_1 - i_2) = 0$$

$$2(i_2 - i_1) + 2i_2 + 2(i_2 + 1) = 0$$

$$i_1 = 4i_a \quad i_1 = -8A$$

$$i_2 = i_a - 1 \quad i_2 = -3A$$

$$2(i_2 - i_1) + 2i_2 + 2(i_2 + 1) = 0$$

$$6i_2 - 2i_1 + 2 = 0$$

$$6(i_a - 1) - 2(4i_a) + 2 = 0$$

$$-2i_a - 4 = 0 \quad \sim i_a = -2A$$